

# Random Variable and Probability Distribution

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# Outline

- ① Random Variable
- ② Probability Distribution
- ③ Cumulative Distribution Function (CDF)
- ④ Mathematical Expectation and Variance

# Random Variable

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# Introduction

A variable which takes numerical values resulting from random experiments, is called a random variable (RV).

- There is a probability associated with each possible values
- Random variables are denoted by capital letters such as  $X, Y, Z$  etc.
- Possible values are denoted by small letters such as  $x, y, z$  etc.
- Example:
  - Height of students
  - Number of heads when tossing a coin three times

# Types of Random Variable

- **Discrete random variable:** A random variable defined over a discrete sample space
  - Number of students in a class, sample space,  $S = \{0, 1, 2, \dots, \infty\}$
  - Number of correct answers in among 50 questions,  $S = \{0, 1, 2, \dots, 50\}$
- **Continuous random variable:** A random variable defined over a continuous sample space
  - Monthly income,  $S = \{x : 0 \leq x < \infty\}$
  - Monthly profit,  $S = \{x : -\infty < x < \infty\}$

# Probability Distribution

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# Probability Distribution

- Probability distribution means the distribution of the probabilities among the different values of a random variable
- For example, when tossing a coin twice, the sample space,  $S = \{HH, HT, TH, TT\}$
- Let an RV,  $X$  = number of heads in two coin tosses, then  $X$  can take the values: 0, 1, 2
- Then the probability of each value of  $X$ :

| $x$ | $P(X = x)$ |
|-----|------------|
| 0   | 1/4        |
| 1   | 2/4        |
| 2   | 1/4        |

# Types of Probability Distributions

Depending on the type of variable, distributions are of two types:

- **Discrete probability distribution:** probability distribution of a discrete random variable
- **Continuous probability distribution:** probability distribution of a continuous random variable

# Discrete probability distribution

- The probability distribution of a discrete random variable is a table, graph, formula, or other device used to specify all possible values of a discrete random variable along with their respective probabilities
- If we let the discrete probability distribution be represented by the function  $p(x)$ , then  $p(x) = P(X = x)$  is the probability of the discrete random variable  $X$  to assume a value  $x$
- $p(x)$  is called a probability mass function (PMF)

# Probability Mass Function (PMF)

A probability mass function gives the probability of each of the value of a discrete random variable.

A function,  $p(x)$ , of a discrete random variable  $X$  will be called a PMF, if and only if all the following conditions are satisfied:

①  $p(x) \geq 0; \forall x$

②  $\sum_x p(x) = 1$

The probability for a value  $a$  of the random variable  $X$  is written as  $p(a)$ , i.e.,  $p(a) = P(X = a)$ .

## Example: PMF

In page 3, the probability mass function of the variable  $X$  representing the number of heads when a coin is tossed twice is given. It is an example of a PMF.

| $x$ | $P(X = x)$ |
|-----|------------|
| 0   | 1/4        |
| 1   | 2/4        |
| 2   | 1/4        |

# Probability Density Function (PDF)

The probability distribution of a continuous random variable is called continuous probability distribution. The probabilities are given by the probability density function (PDF) of that distribution.

A function  $f(x)$  of a continuous random variable  $X$  is called the PDF of  $X$  if and only if the following conditions are satisfied:

- ①  $f(x) \geq 0; \forall x$
- ②  $\int_x f(x)dx = 1$

Here,  $\int_x f(x)dx$  means integration of  $f(x)$  over the whole sample space of  $X$ . Sometimes, it's also written as  $\int_{-\infty}^{\infty} f(x)dx$ .

In the continuous case, the probability that  $X \in [a, b]$  is found by integrating the PDF as:  $P(a \leq X \leq b) = \int_a^b f(x)dx$ .

## PDF: Example

Suppose  $X$  is a continuous random variable whose density function is given by:

$$f(x) = \begin{cases} C(4x - 2x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

- 1 What is the value of  $C$ ?
- 2 Find  $P(X > 1)$ .

# PDF: Example Solution

1

$$\begin{aligned}\int_x f(x)dx &= 1 \\ \Rightarrow \int_0^2 C(4x - 2x^2)dx &= 1 \\ \Rightarrow C[4x^2/2 - 2x^3/3]_0^2 &= 1 \\ \Rightarrow C &= 3/8\end{aligned}$$

2

$$P(X > 1) = \int_1^2 \frac{3}{8}(4x - 2x^2) = 1/2$$

# Probability at a Single Point for Continuous Random Variables

For a continuous RV, probability over an interval is:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

At a single point, the interval collapses to zero width, so:

$$P(X = a) = \int_a^a f(x) dx = 0$$

Since  $P(X = a) = 0$ , adding or removing a boundary point never changes probability:

- $P(X \leq a) = P(X < a)$     and     $P(X \geq a) = P(X > a)$
- $P(a \leq X \leq b) = P(a < X < b)$

# Cumulative Distribution Function (CDF)

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# Cumulative Distribution Function (CDF)

The cumulative distribution function (CDF) gives the probability that a random variable  $X$  is lower than some value  $x$ . Sometimes, it is also called “distribution function”. It is denoted by  $F(x)$ .

- For discrete random variable,  $F(x) = P(X \leq x) = \sum_{k=-\infty}^x p(k)$
- For continuous case,  $F(x) = P(X < x) = P(X \leq x) = \int_{-\infty}^x f(x)dx$
- It is a non-decreasing function, the largest value  $F(x)$  can take is 1

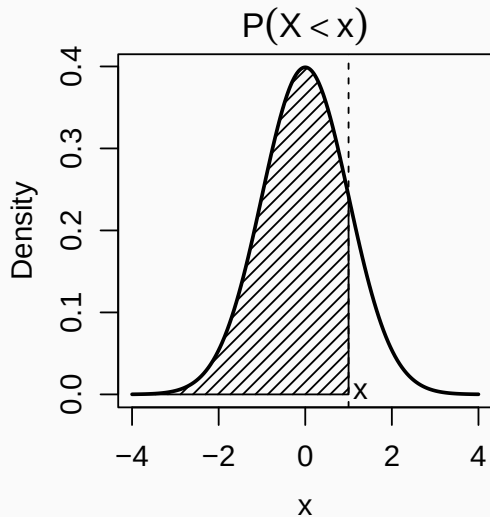
Here, summing/integrating from  $-\infty$  means that the lower limit of the sum/integral is the smallest value in the domain of the random variable  $X$ .

# Graphical Representation of CDF

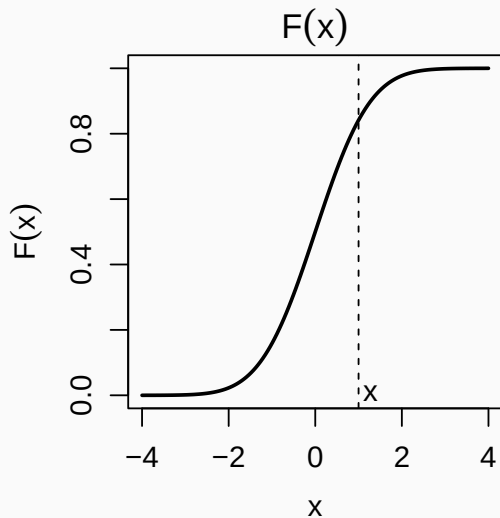
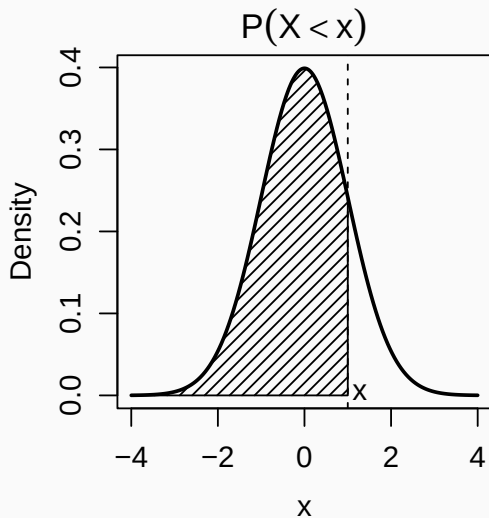
In the plot,  $X$  is a continuous random variable, it is “normally” distributed (bell shaped curve).

The shaded region highlights the probability that the random variable  $X$  is less than some value  $x$  (in the plot, this value is shown to be 1 as an example).

Therefore, the highlighted area is the probability that  $X$  is less than 1.



## Graphical Representation of CDF (cont.)



## CDF: Example 1

In page 3, the probability mass function of the variable  $X$  representing the number of heads when a coin is tossed twice is given. Below, its CDF is also given:

| $x$ | $P(X = x)$ | $P(X \leq x)$ |
|-----|------------|---------------|
| 0   | 1/4        | 1/4           |
| 1   | 2/4        | 3/4           |
| 2   | 1/4        | 1             |

## CDF: Example 2

Suppose  $X$  is a continuous random variable whose density function is given by:

$$f(x) = \begin{cases} \frac{3}{8}(4x - 2x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find the CDF of  $X$ :

$$\begin{aligned} F(x) &= \int_0^x f(x)dx \\ &= \int_0^x \frac{3}{8}(4x - 2x^2)dx \\ &= \frac{3}{8}[(4x^2/2 - 2x^3/3)]_0^x = \frac{3}{8}[(2x^2 - 2x^3/3) - 0] = \frac{3}{8}(2x^2 - 2x^3/3) \end{aligned}$$

# Finding Probability from CDF

## Continuous RV:

- $P(X \leq a) = F(a)$
- $P(X > a) = 1 - F(a)$
- $P(a < X < b) = F(b) - F(a)$

Since  $P(X = a) = 0$ , the  $\leq$  and  $<$  signs are interchangeable for continuous RVs.

## Discrete RV:

- $P(X \leq a) = F(a)$
- $P(X < a) = F(a) - P(X = a)$
- $P(X > a) = 1 - F(a)$
- $P(a < X \leq b) = F(b) - F(a)$
- $P(a \leq X \leq b) = F(b) - F(a) + P(X = a)$

## Example: Finding Probability from CDF of a Continuous RV

Suppose the CDF of a random variable  $X$  is

$$F(x) = x^2 \quad \text{for } 0 \leq x \leq 1$$

Find  $P(0.2 < X < 0.6)$  and  $P(0.2 \leq X \leq 0.6)$ .

Using the CDF rule:

$$\begin{aligned} P(0.2 < X < 0.6) &= F(0.6) - F(0.2) = (0.6)^2 - (0.2)^2 \\ &= 0.36 - 0.04 = 0.32. \end{aligned}$$

Since  $X$  is continuous,  $P(X = a) = 0$ .

Therefore,  $P(0.2 \leq X \leq 0.6) = 0.32$ .

## Example: Finding Probability from CDF of a Discrete RV

Suppose the CDF of a random variable  $X$  satisfies:  $F(2) = 0.35$ ,  $F(5) = 0.80$  and  $P(X = 2) = 0.10$ . Find  $P(2 < X \leq 5)$  and  $P(2 \leq X \leq 5)$ .

First,

$$\begin{aligned}P(2 < X \leq 5) &= F(5) - F(2) \\&= 0.80 - 0.35 = 0.45.\end{aligned}$$

Next,

$$\begin{aligned}P(2 \leq X \leq 5) &= F(5) - F(2) + P(X = 2) \\&= 0.45 + 0.10 = 0.55.\end{aligned}$$

# Mathematical Expectation and Variance

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# Mathematical Expectation

Expectation of an RV,  $X$  is the weighted average of the possible values of  $X$ . The weight of each value of  $X$  is its corresponding probability.

- When  $X$  is discrete:  $E(X) = \sum_x x \cdot p(x)$
- When continuous:  $E(X) = \int_x x \cdot f(x) dx$
- The sum/integral is over all values of  $X$
- It is often denoted by  $\mu$ , that is:  $E(X) = \mu$
- It is also known as expected value or population mean

# Variance of a Random Variable

The variance of an RV,  $X$  measures the dispersion in the values of  $X$ . It is denoted by  $\text{Var}(X)$  or  $V(X)$ .

$$\begin{aligned} V(X) &= E[(X - E(X))^2] \\ &= E[X^2 - 2X \cdot E(X) + [E(X)]^2] \\ &= E(X^2) - 2E(X) \cdot E(X) + [E(X)]^2 \\ &= E(X^2) - 2[E(X)]^2 + [E(X)]^2 \\ &= E(X^2) - [E(X)]^2 \end{aligned}$$

Here,  $E(X^2)$  is given by  $\sum_x x^2 \cdot p(x)$  or  $\int_x x^2 \cdot f(x) dx$  for discrete and continuous case respectively.

# Expectation and Variance: Example 1

In page 3, the PMF of the random variable  $X$  representing the number of heads in two throws of a fair coin is given.

$X$  can take the values 0, 1 and 2 with probabilities  $1/4$ ,  $1/2$  and  $1/4$  respectively.

The expected value of  $X$  is:

$$E(X) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 0 + \frac{1}{2} + \frac{1}{2} = 1$$

The variance of  $X$  is:

$$\begin{aligned} V(X) &= E(X^2) - [E(X)]^2 \\ &= \left[ 0^2 \cdot \frac{1}{4} + 1^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{4} \right] - 1^2 \\ &= \left[ 0 + \frac{1}{2} + 1 \right] - 1 = \frac{1}{2} \end{aligned}$$

## Expectation and Variance: Example 2

Find the expected value and the variance of the RV  $X$  whose PDF is given in page 16.

$$\begin{aligned} E(X) &= \int_0^2 x \cdot \frac{3}{8}(4x - 2x^2)dx \\ &= \frac{3}{8} \int_0^2 (4x^2 - 2x^3)dx \\ &= \frac{3}{8} [4x^3/3 - 2x^4/4]_0^2 \\ &= \frac{3}{8} [4 \times 2^3/3 - 2 \times 2^4/4 - 0] = 1 \end{aligned}$$

Finding the variance is left as an exercise.

# Properties of Expectation and Variance

Let  $a$ ,  $b$  and  $C$  be some constants, and  $X$  and  $Y$  be two independent random variables.

$$\textcircled{1} \ E(c) = c$$

$$\textcircled{2} \ E(cX) = c E(X)$$

$$\textcircled{3} \ E(c + X) = c + E(X)$$

$$\textcircled{4} \ E(aX + bY) = a E(X) + b E(Y)$$

$$\textcircled{1} \ V(c) = 0$$

$$\textcircled{2} \ V(cX) = c^2 V(X)$$

$$\textcircled{3} \ V(c + X) = V(X)$$

$$\textcircled{4} \ V(aX + bY) = a^2 V(X) + b^2 V(Y)$$

If  $X$  and  $Y$  are dependent:

$$V(aX + bY) = a^2 V(X) + b^2 V(Y) + 2ab \text{CoV}(X, Y).$$

Here,  $\text{CoV}(X, Y)$  is the covariance of  $X$  and  $Y$ . More on covariance in a future lecture.

# Properties of Expectation and Variance: Example 1

Suppose  $X$  is a random variable with  $E(X) = 5$  and  $V(X) = 2$ .

- $E(2X) = 2E(X) = 2 \times 5 = 10$
- $V(2X) = 2^2 V(X) = 4 \times 2 = 8$
- $E(2X - 3) = 2E(X) - 3 = 2 \times 5 - 3 = 10 - 3 = 7$
- $V(2X - 3) = 2^2 V(X) = 4 \times 2 = 8$

**Thank you.**

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